STAGE 2

Breadth Diagnostic: Where Is the Ceiling the Binding Constraint?

S019D — Solver fixed (justified by Stage 1). Budget spent on embedding breadth and task breadth.

SHARED WITH STAGE 1

Same ZCTAs (~32k), same embedding code, same spatial holdout

County-holdout evaluation. 5-fold CV. Identical feature pipeline.

FIXED (JUSTIFIED BY STAGE 1: SOLVER-INVARIANT)

1 solver: HistGBDT — isolates embedding effect from solver noise

SCALED UP IN STAGE 2

6 embedding families (3 core + 3 extended)

- PCA32 (no spatial bias)
- Spatial Lag (neighbor)
- GraphSAGE (learned)
- Geo-spatial (context)
- Domain (all 106 cols)
- Noisy control (floor)

27 CONUS targets spanning 5.3x difficulty range

- Health (21)
- Socioeconomic (3)
- Environmental (3)

6 embeddings x 27 targets x 1 solver x 5 folds = 810 fits

What Stage 2 answers

D.1 Does aggregate reporting hide task structure?

N-ceiling ranges 0.084 to 0.444 (5.3x). A single mean R-squared is dominated by easy tasks. Per-task ceiling exposes which tasks are genuinely hard vs under-featured.

D.2 Where is switching embeddings unproductive?

Tasks where 5-family spread is small relative to their ceiling are embedding-invariant: better features, not better architectures, is the productive intervention.

Routing Which feature family wins where?

Domain features wins 16/27 (brute feature breadth).
GNN wins 7/27 (learned spatial structure matters).
Geo-spatial wins 4/27 (geographic context decisive).
Tells you WHERE to invest next.

Stage 2 finding: N-ceiling spectrum (0.084-0.444) is the dominant evaluation axis. Solver spread (0.041) is 9x smaller. The ceiling is the binding constraint.